Intelligent Waste Segregation Using Computer Vision

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1. Abstract

Every piece of waste has its place. This project presents an Intelligent Waste Sorting System that uses computer vision to classify waste into eight categories: cardboard, glass, metal, paper, plastic, vegetation, food organics, and textile trash. By enabling accurate segregation, the system reduces landfill burden, supports recycling, and promotes sustainable resource management, paving the way for a cleaner and greener future.

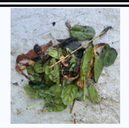
2. Introduction

Improper waste segregation leads to increased landfill burden, inefficient recycling, and higher management costs. This project employs a deep learning–powered mobile application to instantly classify waste into eight categories, providing real-time guidance for proper disposal, reducing environmental impact, and lowering waste management costs.

3. Methodology

DATA COLLECTION → DATA PREPROCESSING → UPLOAD DATASET TO LANDING LENS → MODEL TRAINING → MODEL EVALUATION → RESULT ANALYSIS

4. Input / Output

Test Image	Actual	Predicted	Result
	Glass	Glass	✓
	Plastic	Metal	✗
	Metal	Metal	✓
	Food Organics	Vegetation	✗
	Food Organics	Food Organics	✓

5. Result

The model attained 100% accuracy on the training set, 98% on the validation set, and 94% on the test set, demonstrating excellent learning and strong generalization to unseen data.

6. Conclusion

Instant waste categorization enables households, schools, public spaces, and recycling programs to streamline disposal. Real-time classification enhances recycling efficiency and showcases AI's potential for practical, cost-effective environmental solutions.

7. Steps to use the Application

Step 1 : Take a Sample Photo of any trash material like the ones shown below.

Step 2: Scan the QR code.

Step 3: Upload the picture from gallery (or Take a live photo)

Step 4: Click "Use Photo" and get your result.

Mixed Trash Recycling Crash

Scan & Sort in a Flash !

8. References

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2. Prakash, S., & Kumar, A. "Smart Waste Management Using IoT and AI." Journal of Cleaner Production, 2020.
3. Landing AI. LandingLens Support Documentation. Official Website
4. Kaggle - Real Waste Dataset Kaggle Dataset